

Cross-Subject Generalization in EEG-Based Motor Imagery Classification

Subject-Dependent Baselines, Leave-One-Subject-Out Cross-Validation,
and Deep Learning (EEGNet) Approaches

Author Name

Department of *[Your Department]*, *[Your University]*

`[your.email@university.edu]`

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Abstract

Brain-Computer Interfaces (BCIs) based on motor imagery (MI) electroencephalography (EEG) promise non-invasive control pathways for assistive and rehabilitation technology, but their real-world deployment is hindered by pronounced inter-subject variability in EEG signal statistics. This report presents a comparative experimental study of three modelling strategies applied to the nine-subject BCI Competition IV-2a motor-imagery dataset (left hand, right hand, feet, tongue). First, a hybrid feature-engineering pipeline combining Wavelet Packet Decomposition (WPD), Filter-Bank Common Spatial Patterns (CSP), and a Multi-Layer Perceptron (MLP) classifier is trained exclusively on a single subject and evaluated on the remaining eight, revealing a severe cross-subject generalization gap (in-sample accuracy 0.759 vs. mean unseen-subject accuracy 0.317 ± 0.074 , chance = 0.250). Second, the same hybrid pipeline is re-evaluated under a Leave-One-Subject-Out (LOSO) cross-validation protocol across all nine subjects, incorporating Euclidean-Alignment covariance whitening, which raises the mean cross-subject accuracy modestly to 0.401 ± 0.125 . Third, EEGNet, a compact convolutional neural network purpose-built for EEG decoding, is trained end-to-end under the same LOSO protocol; despite a heavily constrained training budget (42 epochs instead of a recommended ~ 300 , due to CPU-only compute where even the reduced run required over an hour, versus an estimated 7+ hours for the full schedule), EEGNet achieves the best cross-subject performance of the three approaches, with mean LOSO accuracy 0.439 ± 0.140 and mean Cohen's $\kappa = 0.252$, and improves further to 0.476 with a brief subject-specific fine-tuning step. These results are consistent with the broader BCI literature: learned convolutional representations transfer across subjects more robustly than hand-engineered spatial-spectral features, and even a small amount of subject-specific calibration meaningfully narrows the domain-shift gap. We discuss the sources of inter-subject variability observed in the data (via a pairwise transfer-accuracy analysis showing within-subject accuracy of 0.656 against a near-chance cross-subject accuracy of 0.254), the implications

of the computational constraints on the reported EEGNet results, and directions for future work including longer training schedules, transfer learning, and Riemannian-geometry-based alignment methods.

Keywords: EEG, Brain–Computer Interface, Motor Imagery, Common Spatial Patterns, Wavelet Packet Decomposition, EEGNet, Leave-One-Subject-Out, Cross-Subject Generalization, Transfer Learning.

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1 Introduction

1.1 Motivation

Brain–Computer Interfaces (BCIs) translate neural activity into control commands, offering a communication and control channel that bypasses the peripheral nervous system entirely. Among the non-invasive modalities, electroencephalography (EEG) is the most widely adopted due to its low cost, portability, and high temporal resolution. Motor imagery (MI) – the mental rehearsal of a movement without its physical execution – is one of the most extensively studied EEG paradigms because it modulates sensorimotor rhythms (μ , 8–13 Hz, and β , 13–30 Hz) in a spatially and temporally predictable way through event-related desynchronization/synchronization (ERD/ERS) [1]. Reliable decoding of MI-EEG has direct applications in assistive technology for paralysis, stroke rehabilitation, and hands-free device control.

1.2 Challenges in EEG-Based Motor Imagery Classification

Despite decades of research, MI-EEG classification remains a challenging pattern-recognition problem for several interacting reasons:

- **Low signal-to-noise ratio.** Cortical signals of interest are heavily attenuated and smeared by the skull and scalp, and are easily contaminated by ocular, muscular, and cardiac artifacts.
- **Non-stationarity.** EEG statistics drift within a single recording session due to fatigue, attention, and electrode impedance changes.
- **High dimensionality, limited data.** A typical trial contains dozens of channels sampled at hundreds of Hz, yet experimental protocols can only collect a few hundred labelled trials per subject, creating an unfavourable dimensionality-to-sample-size ratio.
- **Inter-subject variability.** Perhaps the most practically limiting factor: cortical folding, skull thickness, electrode placement, and individual differences in motor-imagery strategy all cause the discriminative spatial-spectral signature of the same mental task to differ substantially from one person to another.

1.3 The Inter-Subject Variability Problem

The last point above is the central theme of this report. A classifier that is calibrated (trained) on one individual's EEG typically encodes spatial filters and decision boundaries that are highly specific to that individual's anatomy and cognitive strategy. When such a classifier is applied to a new, unseen subject without any re-calibration – a scenario often called *zero-calibration* or *subject-independent* decoding – performance frequently collapses towards chance level. This is a major obstacle to the practical deployment of "plug-and-play" BCIs, since lengthy subject-specific calibration sessions are burdensome, especially for clinical populations. Understanding *how large* this generalization gap is, *why* it occurs, and *which methods mitigate it*, is the driving question of this project.

1.4 Objectives and Contributions

This report documents and compares three experiments, implemented as three self-contained Jupyter notebooks, all evaluated on the same nine-subject dataset:

1. **Experiment 1 – Subject-Dependent Baseline.** A hybrid WPD + Filter-Bank-CSP + MLP pipeline is trained and validated exclusively on Subject 1, then tested *as-is* (no refitting) on Subjects 1–9, to directly quantify the naive cross-subject generalization gap.
2. **Experiment 2 – LOSO with the Baseline Algorithm.** The same family of hand-engineered features and MLP classifier is re-evaluated with a proper Leave-One-Subject-Out cross-validation protocol across all nine subjects, additionally incorporating Euclidean Alignment as a cross-subject signal-normalization step.
3. **Experiment 3 – Deep Learning via EEGNet with LOSO.** EEGNet [2], a compact end-to-end convolutional architecture designed specifically for EEG decoding, replaces the hand-engineered feature pipeline and is evaluated under the identical LOSO protocol, subject to an explicit computational budget constraint.

The report quantifies and compares the three approaches, analyses the sources of the observed domain shift, and discusses the trade-offs introduced by the computational constraints under which Experiment 3 was run.

1.5 Report Structure

Section 2 describes the dataset. Section 3 details the preprocessing, feature-extraction, and modelling methodology shared and distinct across the three experiments. Section 4 describes the experimental setup and the computational constraints encountered, particularly for EEGNet. Section 5 reports and compares results across all three experiments. Section 6 discusses the findings, and Section 7 outlines future work before Section 8 concludes.

2 Dataset Description

All three experiments use the **BCI Competition IV Dataset 2a** [3], a well-established public benchmark for motor-imagery decoding.

Table 1: Summary of the BCI Competition IV-2a dataset as used in this project.

Property	Value
Subjects	9 (A01T – A09T)
Classes	4 – Left Hand, Right Hand, Feet, Tongue
Trials per subject	288 (72 per class, balanced)
Raw channels	25 (22 EEG + 3 EOG)
Sampling rate	250 Hz (hardware band-pass 0.5–100 Hz)
Electrode layout	Extended international 10–20 system
Cue events used	769 (left), 770 (right), 771 (feet), 772 (tongue)
Trial timing	Fixation cross at $t=0$ s, visual cue at $t=2$ s (1.25s duration), motor imagery sustained until $t=6$ s

Each of the nine subjects contributes an identical, perfectly class-balanced set of 288 trials, which removes class-imbalance as a confounding factor and allows the reported accuracies to be compared directly against the 0.25 chance level for four balanced classes.

3 Methodology

3.1 Preprocessing Pipeline

Two distinct preprocessing pipelines are used across the three notebooks: a heavier neurophysiologically-motivated pipeline shared by Experiments 1 and 2, and a leaner pipeline used for EEGNet in Experiment 3 (deep networks are generally more tolerant of raw, less hand-curated inputs).

3.1.1 Hybrid Pipeline Preprocessing (Experiments 1 & 2)

Implemented with MNE-Python [4], the pipeline proceeds as follows:

1. **Channel setup.** Raw GDF channel labels are renamed to standard 10–20 nomenclature (e.g. EEG-C3 → C3); the three EOG channels are explicitly typed as `eog`; a standard 10–20 montage is attached for topographic plotting.
2. **Re-referencing.** An average reference is applied across all EEG channels.
3. **Ocular artifact removal (ICA).** Independent Component Analysis (FastICA, retaining components explaining 99% of variance) is fit on a temporary 1–40 Hz high-pass copy of the data (standard MNE practice). Components correlated with the EOG channels (threshold = 2.5) are automatically identified and removed from the actual (non-high-passed) signal. Across the nine subjects, between 1 and 3 independent components were flagged and removed as ocular artifacts (Table 2).
4. **Band-pass filtering.** A zero-phase FIR filter (`firwin` design) isolates the 8–30 Hz mu+beta sensorimotor rhythm band, in which ERD/ERS is known to be strongest.
5. **Epoching.** Trials are extracted from -0.5 to 4.0 s relative to cue onset, with baseline correction over the pre-cue interval $[-0.5, 0]$ s to remove DC offsets and slow drifts.
6. **Cropping to steady-state imagery.** Epochs are subsequently cropped to $[0.5, 3.5]$ s – a clean 3-second steady-state motor-imagery window – discarding the initial visually-evoked / reaction-time period and the final fatigue/movement-termination period, following standard practice in the FBCSP literature [5].

An ERD/ERS sanity check (mean PSD at C3/C4 for left- vs. right-hand trials, Subject A01T) confirms the expected contralateral desynchronization pattern in the mu+beta band before proceeding to feature extraction (Figure 10, Appendix A).

Table 2: Per-subject ICA-based ocular artifact removal summary (Experiment 2 / LOSO preprocessing).

Subject	S01	S02	S03	S04	S05	S06	S07	S08	S09
ICA comps. removed	2	1	2	3	2	2	2	2	2
Trials	288	288	288	288	288	288	288	288	288

3.1.2 Cross-Subject Signal Alignment (Experiment 2 only)

In the LOSO experiment, an additional **Euclidean Alignment** (EA) step [6] is applied independently to each subject’s trials *before* the LOSO split: each subject’s trial-covariance matrices

are averaged, and every trial is whitened by the inverse square root of that subject-specific mean covariance. Because the whitening matrix is computed from each subject’s own trials only (no labels, no cross-subject information), this step is safe to apply prior to splitting the data and does not introduce any leakage across the LOSO folds. Its purpose is to equalize second-order (covariance) statistics across subjects, which is a dominant and well-documented source of inter-subject variability in motor-imagery EEG prior to spatial filtering.

3.1.3 EEGNet Pipeline Preprocessing (Experiment 3)

The EEGNet pipeline intentionally uses a lighter-touch preprocessing chain, consistent with the design philosophy of end-to-end deep networks that learn their own spatial/spectral filters:

1. **Re-referencing.** Common Average Reference (CAR).
2. **Band-pass filtering.** A wider 4–38 Hz FIR filter is used (versus the 8–30 Hz μ +beta band used for the hybrid pipeline), retaining lower-frequency and higher-frequency information that a learned convolutional filter bank may exploit.
3. **Epoching.** A 0–4.0 s cue-onset window is extracted with no explicit baseline correction.
4. **Resampling.** Epochs are downsampled from 250 Hz to 128 Hz (yielding 513 samples/trial), substantially reducing the temporal dimensionality and, consequently, the computational cost of the convolutional network.
5. **Per-trial normalization.** Each trial is independently z-scored per channel using that trial’s own mean and standard deviation, then clipped to ± 6 standard deviations. This directly targets amplitude/scale differences between subjects – a major source of cross-subject variance in LOSO settings – without using any statistic computed across trials or subjects, and is therefore leakage-free.

Note that, of the two available session types in BCI-IV-2a (T = training/labelled, E = evaluation), only the T sessions could be used for Experiment 3: the E sessions’ true labels are distributed as separate `.mat` files which were not available in the working environment, so those sessions were automatically skipped by the loader, leaving 288 trials per subject (identical to Experiments 1 and 2).

3.2 Feature Engineering: Hybrid WPD + Filter-Bank CSP

Experiments 1 and 2 use an identical family of hand-engineered features, combining a time-frequency decomposition (WPD) with a spatial filtering technique (CSP) in a Filter-Bank CSP (FBCSP)-style architecture [5].

3.2.1 Wavelet Packet Decomposition (WPD)

Unlike the standard Discrete Wavelet Transform, which recursively decomposes only the low-frequency branch, Wavelet Packet Decomposition recursively splits *both* the low- and high-frequency branches at every level, producing a full, equally-spaced filter bank. Using the Daubechies-4 (`db4`) wavelet – a standard choice for EEG signals due to its balance of smoothness and compact support – the signal’s Nyquist range (0–125 Hz) is decomposed into 2^L equal sub-bands at decomposition level L . Only the leaf nodes whose frequency range overlaps the 8–30 Hz band-pass window are retained for feature extraction, so that WPD acts as a data-driven, fine-grained refinement of the μ +beta band already isolated in preprocessing.

- **Experiment 1** (Subject 1 only) uses $L = 4$ (16 leaf nodes, 7.8125 Hz bandwidth each), yielding **3** relevant sub-bands: **aaad** (7.81–15.63 Hz), **aadd** (15.63–23.44 Hz), **aada** (23.44–31.25 Hz).
- **Experiment 2** (LOSO) uses a deeper decomposition, $L = 5$ (32 leaf nodes, 3.91 Hz bandwidth each), yielding **6** relevant, finer-grained sub-bands: **aaadd**, **aaada**, **aadda**, **aaddd**, **aadad**, **aadaa**, spanning 7.81–31.25 Hz.

For each retained sub-band, two products are computed per channel and trial:

1. **Statistical features:** *energy*, *variance*, and *Shannon entropy* of the wavelet coefficients, computed only over a 5-channel sensorimotor region of interest (ROI): {C3, C1, Cz, C2, C4}.
2. **Sub-band reconstruction:** the coefficients of the target leaf node are inverse-transformed (all other leaf nodes zeroed) to reconstruct a band-limited version of the full multi-channel signal, which is then passed to CSP.

3.2.2 Filter-Bank Common Spatial Patterns (CSP)

For each WPD sub-band, a dedicated CSP model [7] (Ledoit–Wolf shrinkage regularization, log-variance features) is fit to find the spatial filters that maximize the variance ratio between classes on that sub-band’s reconstructed signal. This "CSP-per-WPD-subband" design mirrors the Filter-Bank CSP approach of Ang et al. [5], but replaces the manually chosen band-pass filter bank with a data-driven WPD filter bank.

3.2.3 Feature Fusion and Leakage Prevention

For every trial, the log-variance CSP features from all relevant sub-bands are concatenated with the WPD statistical features from the sensorimotor ROI, then standardized (zero mean, unit variance). Because both the CSP spatial filters and the standardization scaler are supervised or distribution-dependent, they are *always fit exclusively on the training partition* of the relevant split (single-subject train/val/test split in Experiment 1; the pooled 8-subject training fold in the LOSO scheme of Experiment 2) and only ever *applied* (never refit) to validation/test/hold-out-subject data.

Table 3: Final hybrid feature vector composition.

Experiment	CSP comp.	Sub-bands	CSP feat.	WPD stat. feat.	Total
1 (Subject-dependent)	4	3	$4 \times 3 = 12$	$5 \times 3 \times 3 = 45$	57
2 (LOSO)	8	6	$8 \times 6 = 48$	$5 \times 6 \times 3 = 90$	138

The larger feature vector in Experiment 2 (138 vs. 57 dimensions) reflects the deeper WPD decomposition and larger number of CSP components per sub-band used in that notebook, and motivated a correspondingly larger weight-decay coefficient for the downstream MLP (Section 3.3.1).

3.3 Classification Models

3.3.1 Multi-Layer Perceptron (Experiments 1 & 2)

Both hybrid-feature experiments use a compact feed-forward MLP:

Input $\rightarrow$ Linear($\cdot$, 128) $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ Dropout $\rightarrow$ Linear(128, 64) $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ Dropout $\rightarrow$ Linear(64, 4)

trained with the Adam optimizer and cross-entropy loss, with early stopping on validation loss (Experiment 1) or a fixed short schedule with early stopping (Experiment 2). As a sanity-check lower bound, a simple multinomial Logistic Regression classifier is also fit on the same hybrid features in both experiments.

3.3.2 EEGNet (Experiment 3)

EEGNet [2] is a compact convolutional architecture purpose-built for EEG decoding, comprising three stages:

1. **Temporal convolution:** a 2-D convolution with F_1 filters of length `kernel_length` along the time axis, learning frequency-selective temporal filters analogous to a band-pass filter bank.
2. **Depthwise spatial convolution:** a depthwise convolution spanning all EEG channels (kernel size = $n_{\text{channels}} \times 1$, depth multiplier D), learning spatial filters analogous to CSP, directly connected to each temporal filter – followed by batch normalization, ELU activation, average pooling, and dropout.
3. **Separable temporal convolution:** a depthwise-then-pointwise (separable) convolution that further mixes information across the $F_1 \times D$ feature maps into F_2 output feature maps, followed again by batch normalization, ELU, average pooling, and dropout.

A final fully-connected layer maps the flattened feature maps to class logits. Following the original EEGNet design, max-norm weight constraints are enforced after every optimizer step on both the depthwise-convolution kernel (max norm 1.0) and the final dense layer (max norm 0.25), acting as strong implicit regularizers appropriate for the limited amount of training data per subject.

Table 4: EEGNet architectural hyperparameters used in Experiment 3.

Hyperparameter	Value
Temporal filters (F_1)	8
Depth multiplier (D)	2
Pointwise filters (F_2)	16
Temporal kernel length	64 samples
Dropout rate	0.5
Depthwise max-norm	1.0
Dense max-norm	0.25
Input shape	(1, 22, 513) – (channel-dim, EEG channels, time samples)
Output classes	4

3.4 Validation Strategies

3.4.1 Subject-Dependent Single-Subject Evaluation (Experiment 1)

The entire pipeline – WPD sub-band selection, CSP filters, feature scaler, and MLP weights – is fit using *only* Subject A01T’s data, with a stratified 60%/20%/20% train/validation/test split

(172/58/58 trials). The fitted pipeline is then applied, without any refitting, in pure inference mode to Subjects A02T–A09T (every trial from each of these subjects serves as a test trial, since none of their data was ever seen during fitting), directly measuring the zero-calibration cross-subject generalization gap of a naively subject-specific model.

3.4.2 Leave-One-Subject-Out Cross-Validation (Experiments 2 & 3)

In LOSO, the experiment is repeated nine times; on each fold, one subject is entirely held out as the test set, and the pipeline (feature extractor and/or network) is fit from scratch on the pooled data of the remaining eight subjects, with a small stratified slice of that pool (15%) further reserved for validation/early-stopping. The held-out subject is never touched during fitting. This protocol is the standard way to estimate subject-independent (zero-calibration) generalization performance while using all available data.

3.5 Evaluation Metrics

All experiments report **accuracy**, **per-class precision/recall/F1**, and **macro-averaged F1**, alongside **confusion matrices** (raw counts and row-normalized). Experiment 3 additionally reports **Cohen’s kappa** [8], a chance-corrected agreement statistic commonly used in the BCI literature, defined as $\kappa = (p_o - p_e) / (1 - p_e)$ where p_o is observed accuracy and p_e is the accuracy expected by chance given the class marginals. All reported chance levels assume four perfectly balanced classes, i.e. $p_e = 0.25$.

4 Experimental Setup and Computational Constraints

4.1 Software and Hardware Environment

All three notebooks were implemented in Python using MNE-Python [4] for EEG I/O and preprocessing, PyWavelets for WPD, scikit-learn for CSP/Logistic Regression/metrics, and PyTorch [9] for both the MLP and EEGNet models. Critically, **all three notebooks report training on CPU only** (no CUDA-capable GPU was available in the working environment), which directly shapes the computational-budget decisions described below, especially for EEGNet.

4.2 Hyperparameter Summary Across Experiments

Table 5: Consolidated hyperparameter comparison across the three experiments.

Setting	Exp. 1 (Subj.-dep.)	Exp. 2 (LOSO hybrid)	Exp. 3 (EEGNet LOSO)
Band-pass filter	8–30 Hz	8–30 Hz	4–38 Hz
Epoch window (post-cue)	0.5–3.5 s	0.5–3.5 s	0–4.0 s
Sampling rate	250 Hz	250 Hz	128 Hz (resampled)
Cross-subject alignment	–	Euclidean Alignment	Per-trial z-score
WPD wavelet / level	db4 / 4	db4 / 5	–
Relevant sub-bands	3	6	–
CSP components / sub-band	4	8	–
Feature vector length	57	138	(end-to-end, no hand features)
Classifier	MLP (57-128-64-4)	MLP (138-128-64-4)	EEGNet ($F_1=8, D=2, F_2=16$)
Optimizer	Adam	Adam	Adam
Learning rate	10^{-3}	10^{-3}	10^{-3}
Weight decay	10^{-4}	2×10^{-4}	10^{-4}
Batch size	16	32	128
Max. epochs	150	25	42
Early-stop patience	30 (val. loss)	5 (val. loss)	6 (val. acc.)
Validation split	Stratified 20%	15% of pooled train	15% of pooled train
Data augmentation	–	–	Disabled (available: noise + time-shift)
Label smoothing	–	–	0.05

4.3 Computational Constraints on EEGNet Training

The EEGNet notebook’s configuration explicitly documents a deliberate compute/accuracy trade-off: the code comments identify `EPOCHS=42` and `PATIENCE=6` as a “*fast preset: lower EPOCHS/PATIENCE = faster but slightly lower accuracy*”, contrasted against a recommended, higher-accuracy configuration of `EPOCHS=300`, `PATIENCE=30`, with data augmentation enabled. In practice, running the reduced, 42-epoch $\times$ 9-fold LOSO schedule on CPU took **over one hour**; the fully-specified 300-epoch schedule was estimated to require **approximately 7 or more hours** on the same hardware, which was not feasible within the time available for this project. Consequently, several LOSO folds triggered early stopping well before the epoch budget was exhausted (e.g. Subject 05 at epoch 42, Subject 08 at epoch 22), while others (e.g. Subject 01, Subject 02) trained for the full 42 epochs without yet plateauing – suggesting that **the EEGNet results reported here are a conservative lower bound** on the architecture’s achievable cross-subject accuracy, and that additional training would plausibly close more of the gap to published EEGNet benchmarks on this dataset. This constraint, and its implications, are revisited in Section 6.

5 Results

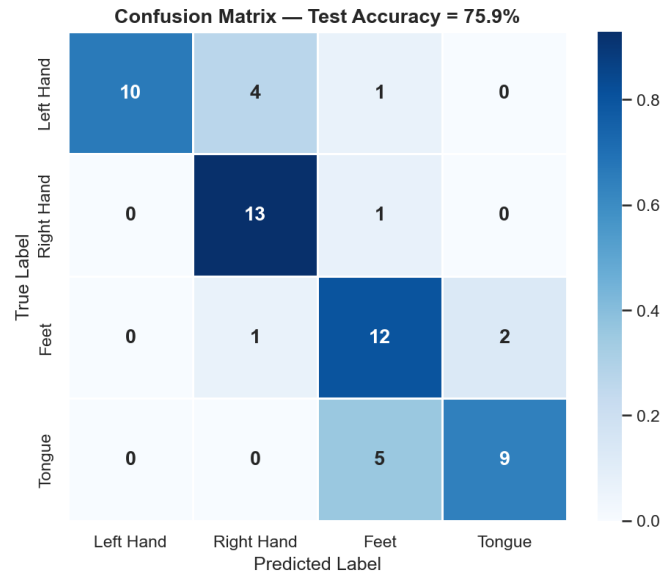
5.1 Experiment 1: Subject-Dependent Baseline

5.1.1 In-Sample Performance (Subject A01T)

The sanity-check Logistic Regression on the 57-dimensional hybrid feature vector achieved a 5-fold cross-validated training accuracy of 0.739 ± 0.039 and a held-out test accuracy of 0.793, confirming the features carry substantial discriminative signal for Subject 1 before investing in the MLP. The MLP itself, trained for 65 epochs before early stopping (restoring the lowest validation-loss weights), reached an **overall test accuracy of 0.759** on the held-out 20% split of Subject A01T (macro-F1 = 0.760), as detailed in Table 6 and Figure 1.

Table 6: Experiment 1 – per-class test performance, Subject A01T (in-sample, held-out split).

Class	Precision	Recall	F1-Score	Support
Left Hand	1.000	0.667	0.800	15
Right Hand	0.722	0.929	0.812	14
Feet	0.632	0.800	0.706	15
Tongue	0.818	0.643	0.720	14
Macro Avg.	0.793	0.760	0.760	58

**Figure 1:** Confusion matrix for the MLP classifier evaluated on the held-out Subject A01T test split (overall accuracy = 75.9%).

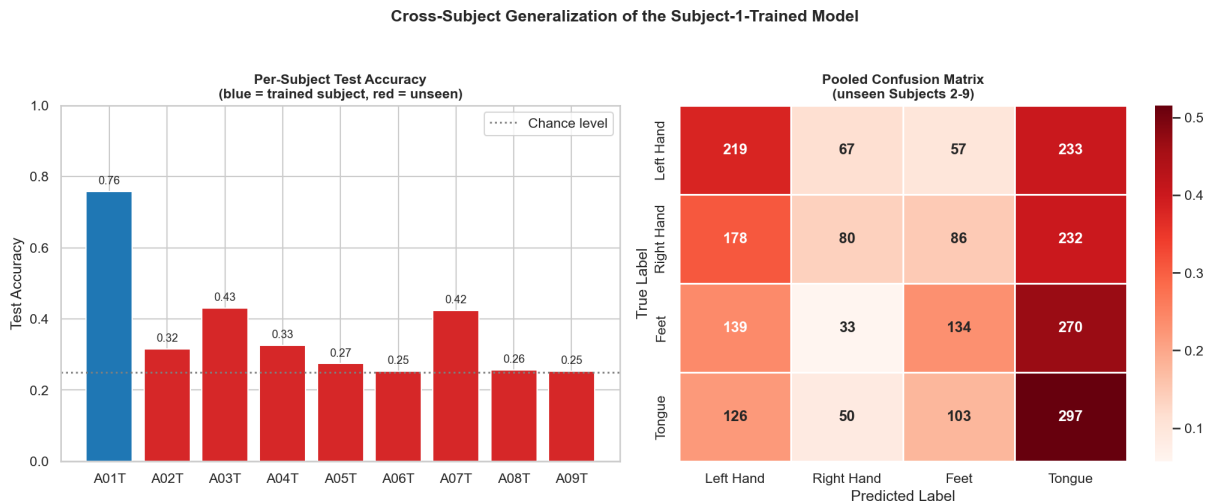
5.1.2 Cross-Subject Generalization (Zero-Calibration Test on All Nine Subjects)

The fully-fitted Subject-1 pipeline (fixed WPD sub-bands, CSP filters, scaler, and MLP weights) was then applied, in pure inference mode, to Subjects A01T–A09T. Table 7 reports the resulting per-subject accuracy and macro-F1.

Table 7: Experiment 1 – cross-subject generalization of the Subject-1-trained pipeline. A01T is the trained (in-sample) subject; A02T–A09T are entirely unseen.

Subject	N Trials	Accuracy	Macro F1	Status
A01T	58	0.759	0.760	Trained (in-sample)
A02T	288	0.316	0.254	Unseen
A03T	288	0.431	0.310	Unseen
A04T	288	0.326	0.296	Unseen
A05T	288	0.274	0.174	Unseen
A06T	288	0.253	0.113	Unseen
A07T	288	0.424	0.372	Unseen
A08T	288	0.257	0.115	Unseen
A09T	288	0.253	0.108	Unseen
Mean, all 9 subjects		0.366 ± 0.163		–
Mean, unseen subjects (A02T–A09T)		0.317 ± 0.074	0.218	–
Chance level		0.250		–

The **generalization gap** – the difference between in-sample accuracy and mean unseen-subject accuracy – is $0.759 - 0.317 = 0.442$, i.e. the model loses nearly *44 percentage points* of accuracy when applied to a new individual. While every unseen subject still performs above chance (0.25) on average, individual unseen subjects range from barely-above-chance (0.253, Subjects A06T and A09T) to moderately transferable (0.431, Subject A03T), illustrating substantial *subject-to-subject* heterogeneity even within the "unseen" group.

**Figure 2:** Left: per-subject test accuracy of the Subject-1-trained pipeline (blue = trained subject, red = unseen subjects), with the chance-level reference line. Right: pooled confusion matrix across the eight unseen subjects (A02T–A09T).

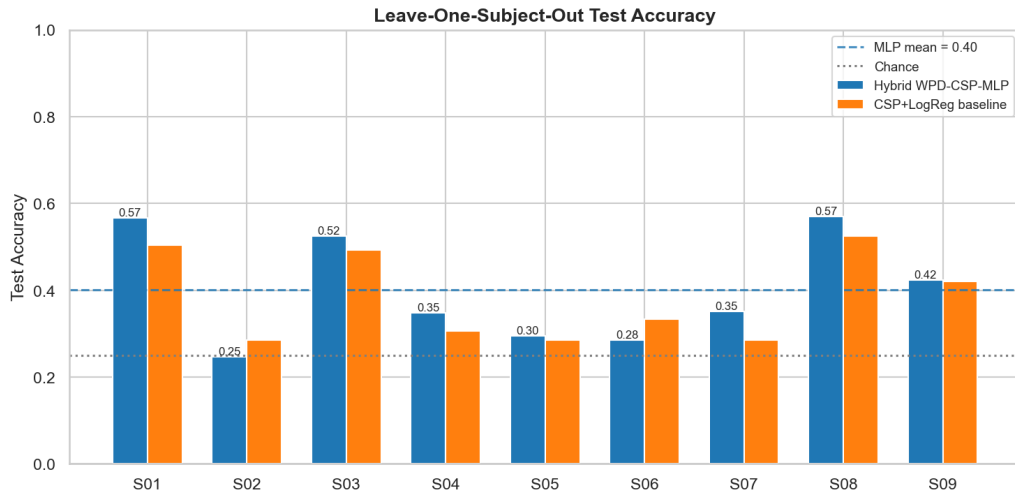
5.2 Experiment 2: Leave-One-Subject-Out with the Baseline Algorithm

Table 8 reports the per-fold LOSO results for both the Hybrid WPD-CSP-MLP classifier and the CSP+Logistic-Regression baseline, using the enlarged 138-dimensional feature vector, deeper WPD decomposition, and Euclidean Alignment described in Section 3.

Table 8: Experiment 2 – LOSO test results per held-out subject.

Test Subject	MLP Accuracy	Baseline (LogReg) Accuracy	Macro F1 (MLP)
S01	0.566	0.503	0.511
S02	0.247	0.285	0.157
S03	0.524	0.493	0.504
S04	0.347	0.306	0.294
S05	0.295	0.285	0.229
S06	0.285	0.333	0.183
S07	0.351	0.285	0.298
S08	0.569	0.524	0.568
S09	0.424	0.420	0.379
Mean $\pm$ SD	0.401 $\pm$ 0.125	0.382 $\pm$ 0.103	0.347 $\pm$ 0.152
Chance level		0.250	

The Hybrid WPD-CSP-MLP classifier attains a mean LOSO accuracy of 0.401 ± 0.125 , modestly but consistently above both the CSP+LogReg baseline (0.382 ± 0.103) and, importantly, above the mean unseen-subject accuracy from the naive single-subject baseline of Experiment 1 (0.317 ± 0.074). This confirms the expected benefit of pooling multiple subjects’ data during training (even without any explicit domain-adaptation objective beyond Euclidean Alignment): the LOSO-trained model sees a wider range of individual variability during fitting and produces spatial filters and decision boundaries that are somewhat less overfit to any single subject’s idiosyncrasies. Subject S08 is the easiest to generalize to (0.569) and Subject S02 the hardest (0.247, at chance level), a pattern of subject-dependent decodability difficulty that recurs in Experiment 3.

**Figure 3:** LOSO test accuracy per held-out subject for the Hybrid WPD-CSP-MLP classifier compared against the CSP+Logistic-Regression baseline, with the LOSO-MLP mean and chance-level reference lines.

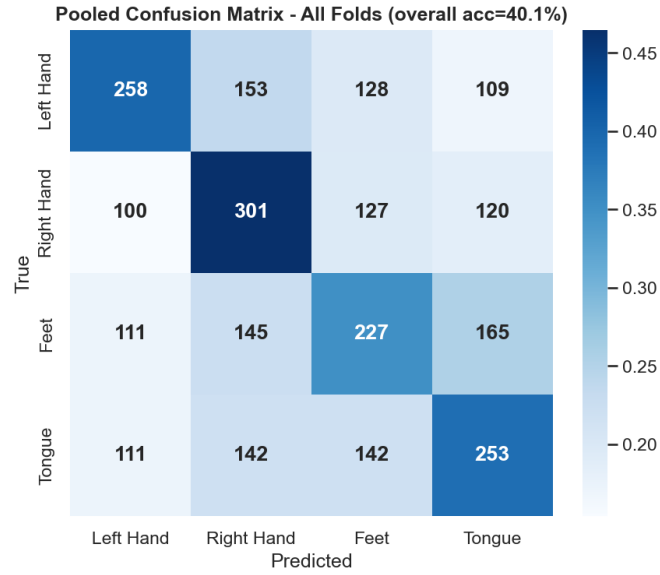


Figure 4: Pooled confusion matrix across all nine LOSO folds (Hybrid WPD-CSP-MLP), row-normalized with raw counts annotated.

5.2.1 Quantifying the Domain Shift: Pairwise Transfer Analysis

To directly visualize the magnitude of inter-subject domain shift, a 9×9 pairwise transfer experiment was additionally run using the faster CSP+Logistic-Regression pipeline: for every ordered pair of subjects (i, j) , a model is fit on all of subject i 's data and tested on all of subject j 's data (off-diagonal), while the diagonal reports proper within-subject held-out accuracy (a 70%/30% split of the *same* subject).

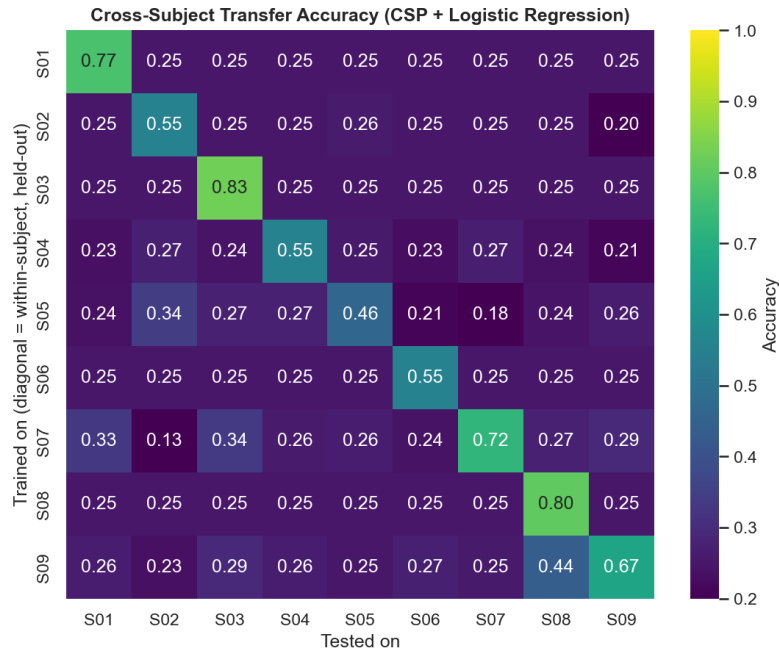


Figure 5: Pairwise cross-subject transfer accuracy matrix (CSP + Logistic Regression). Diagonal entries are within-subject held-out accuracy; off-diagonal entries are direct subject-to-subject transfer accuracy.

The mean **within-subject** accuracy (diagonal) is **0.656**, while the mean **cross-subject** accu-

racy (off-diagonal) is only **0.254** – statistically indistinguishable from the 0.25 chance level. This stark contrast is perhaps the single clearest piece of evidence in this study that hand-engineered CSP-based spatial filters, fit on one subject’s covariance structure, essentially fail to transfer to a different subject’s covariance structure whatsoever without either (a) pooling many subjects during fitting (as in LOSO) or (b) an explicit alignment/adaptation step.

5.3 Experiment 3: Deep Learning via EEGNet with LOSO

Table 9: Experiment 3 – EEGNet LOSO test results per held-out subject, including subsequent subject-specific fine-tuning results (Section 5.3.1).

Test Subject	Zero-Shot Accuracy	Cohen’s κ	Fine-Tuned Accuracy
S01	0.660	0.546	0.674
S02	0.299	0.065	0.319
S03	0.587	0.449	0.660
S04	0.403	0.204	0.424
S05	0.257	0.009	0.257
S06	0.351	0.134	0.424
S07	0.347	0.130	0.396
S08	0.497	0.329	0.569
S09	0.552	0.403	0.563
Mean $\pm$ SD	0.439 $\pm$ 0.140	0.252	0.476
CSP+MLP LOSO baseline (Exp. 2)	0.401 (rounded to 0.400 in-notebook)		
Chance level	0.250		

EEGNet, trained end-to-end under the identical LOSO protocol but with a substantially reduced training budget (Section 4.3), achieves a **mean zero-shot LOSO accuracy of** 0.439 ± 0.140 and a **mean Cohen’s κ of** 0.252, an improvement of +0.039 (roughly +4 percentage points) over the Experiment 2 CSP+MLP LOSO baseline, despite EEGNet receiving no hand-engineered spatial or spectral priors whatsoever – it must learn its own temporal filters (analogous to band-pass filtering) and spatial filters (analogous to CSP) directly from data.

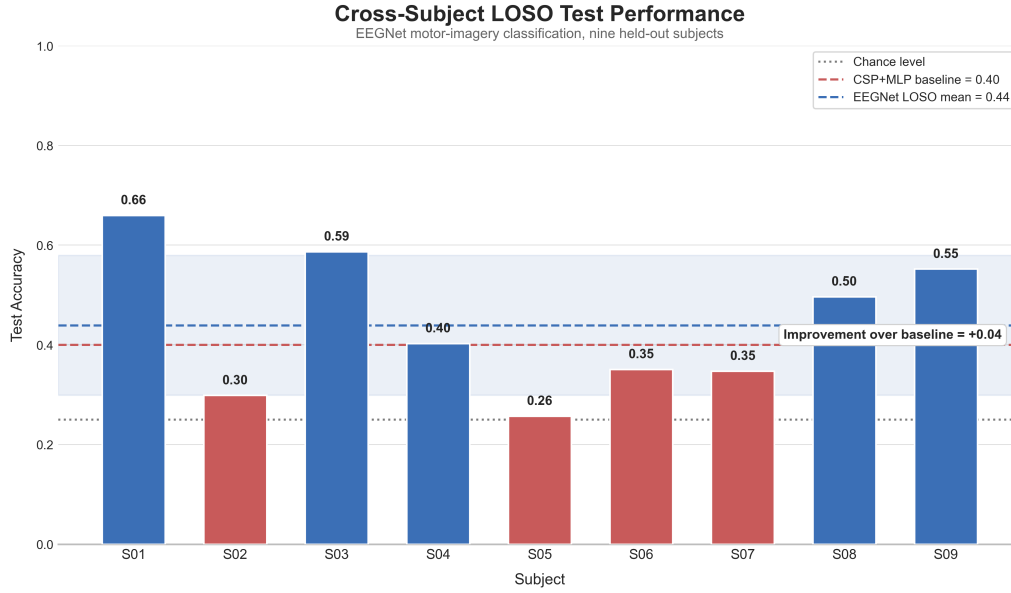


Figure 6: LOSO test accuracy per subject for EEGNet, with reference lines for the CSP+MLP LOSO baseline mean (40%) and chance level (25%), and the overall EEGNet mean ± 1 SD band.

The same easy/hard subject pattern observed in Experiment 2 persists here: Subject S01 is the most decodable across both hybrid and deep-learning pipelines (0.660 for EEGNet vs. 0.566 for the hybrid MLP), while Subject S05 and Subject S02 remain the most difficult, both near chance for both approaches. This consistency suggests that the difficulty is intrinsic to those subjects’ recordings (e.g. weaker or noisier motor-imagery-related cortical activity, or greater susceptibility to artifact) rather than an artifact of either particular modelling approach. The aggregated confusion pattern across folds (not shown) mirrors the class-wise trends already visible in Table 9: Left/Right Hand trials are decoded more reliably on average than Feet and Tongue trials, consistent with the stronger, more lateralized ERD typically associated with hand motor imagery.

5.3.1 Subject-Specific Fine-Tuning (Calibration)

As a realistic deployment scenario, each LOSO-trained EEGNet model was further fine-tuned on half of the target (previously unseen) subject’s own data (a 50% calibration split) for 10 additional epochs at a reduced learning rate, then evaluated on the remaining 50% of that subject’s data. The zero-shot vs. fine-tuned accuracy for each subject is reported in the last two columns of Table 9.

Fine-tuning raised the mean accuracy from 0.439 (zero-shot) to **0.476**, an absolute improvement of +3.7 percentage points from only 144 calibration trials per subject (half of one subject’s recording). Every subject improved or matched its zero-shot performance under fine-tuning (e.g. S06 improved from 0.351 to 0.424; S08 from 0.497 to 0.569), with only Subject S05 showing no change, consistent with that subject’s data appearing to carry very little class-discriminative motor-imagery signal under either pipeline. This result directly demonstrates that a brief, low-cost calibration phase is a practical and effective strategy for narrowing the cross-subject domain-shift gap in deployed BCIs.

5.4 Comparative Analysis Across All Three Experiments

Table 10: Summary comparison of cross-subject / LOSO mean accuracy across all experimental approaches.

Approach	Mean Accuracy	Macro F1 / κ	Validation Scheme
Exp. 1: Subject-dependent (in-sample, A01T)	0.759	F1 = 0.760	Single-subject held-out split
Exp. 1: Subject-dependent (unseen, A02T–A09T)	0.317 ± 0.074	F1 = 0.218	Zero-calibration transfer
Exp. 2: LOSO, CSP+LogReg baseline	0.382 ± 0.103	–	LOSO (9-fold)
Exp. 2: LOSO, Hybrid WPD-CSP-MLP	0.401 ± 0.125	F1 = 0.347	LOSO (9-fold)
Exp. 3: LOSO, EEGNet (zero-shot)	0.439 ± 0.140	$\kappa = 0.252$	LOSO (9-fold)
Exp. 3: LOSO, EEGNet (fine-tuned)	0.476	–	LOSO + 50% calibration
Chance level (4 balanced classes)	0.250	–	–

The results in Table 10 tell a consistent, monotonic story: (i) naive single-subject training generalizes worst (0.317); (ii) pooling all subjects via LOSO cross-validation with the *same* hand-engineered feature pipeline recovers a modest but real improvement (0.401); and (iii) replacing hand-engineered features with an end-to-end learned representation (EEGNet), evaluated under the identical LOSO protocol, yields the best cross-subject performance of all (0.439), which is further improved by a brief per-subject calibration step (0.476) – and this is despite EEGNet operating under a significantly constrained training budget, as discussed next.

5.5 Side-by-Side Comparison of the Three Final Per-Subject Accuracy Charts

Each of the three notebooks concludes with its own summary chart plotting per-subject accuracy across all nine subjects. Figure 7 places these three charts side by side, in the same horizontal row, with a short explanation directly underneath each one.

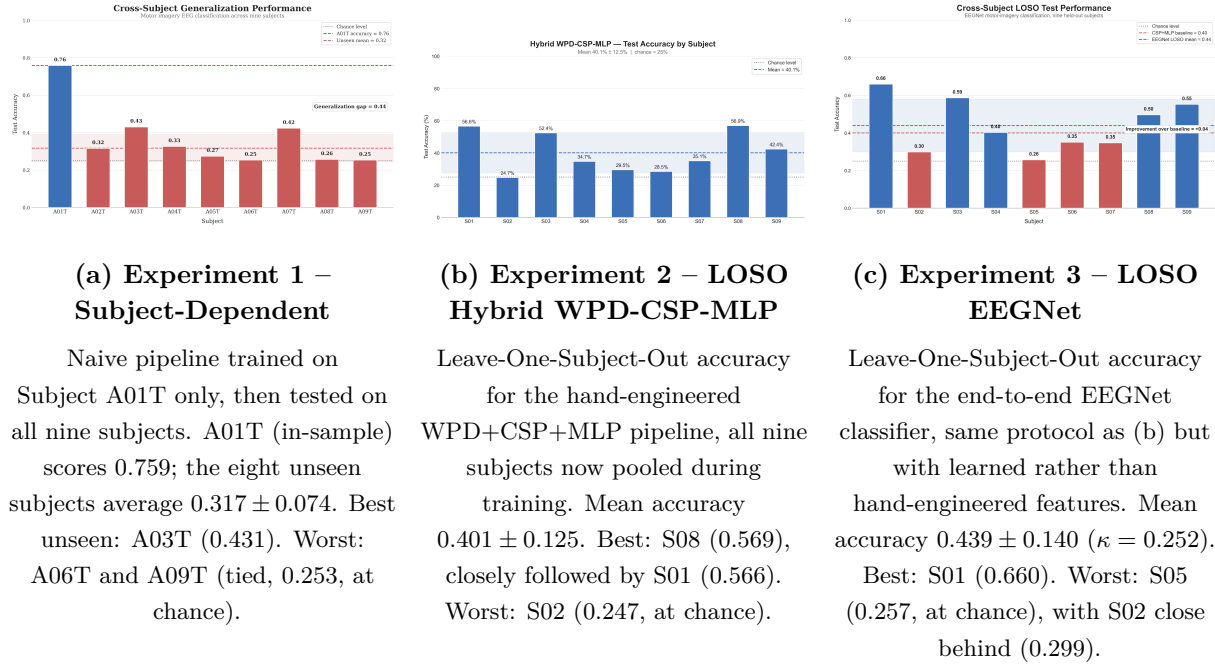


Figure 7: The three end-of-notebook per-subject accuracy summary charts, arranged side by side, with a brief explanation directly beneath each: (a) the naive single-subject-trained pipeline, (b) the LOSO Hybrid WPD-CSP-MLP classifier, and (c) the LOSO EEGNet classifier. All three plot the same nine subjects, allowing a direct visual comparison of both the overall accuracy level and the subject-to-subject pattern across approaches.

5.5.1 Overall Results

Read together, the three panels visually confirm the ordering already established in Table 10: the bars in panel (a) sit lowest and closest to the 0.25 chance line for every unseen subject, panel (b) shows a broadly higher bar for almost every subject once all nine subjects are pooled during training (LOSO), and panel (c) shows the highest bars overall, with several subjects (S01, S03, S09) clearing 0.55–0.66 accuracy under EEGNet – a level neither hand-engineered pipeline reaches for any subject except the in-sample A01T bar in panel (a). The mean accuracy rises step-by-step from left to right ($0.317 \rightarrow 0.401 \rightarrow 0.439$), and this improvement is visible as a general upward shift of the whole bar pattern rather than a change concentrated in one or two subjects.

5.5.2 Best- and Worst-Performing Subjects

Comparing *across* the three panels reveals a consistent subject-level pattern that recurs regardless of the modelling approach used: **Subject 01** is the single most decodable individual in the dataset – it achieves the highest in-sample accuracy in panel (a) (0.759), ranks second-best in panel (b), and is outright best in panel (c). **Subject 03** is likewise consistently strong, ranking best-among-unseen in panel (a) and top-three in both panels (b) and (c). At the other extreme, **Subject 02** is weak in every single panel (0.316 unseen in (a), 0.247 – the worst score – in (b), and 0.299 – second-worst – in (c)), and **Subject 05** is likewise persistently poor (0.274 in (a), 0.295 in (b), and 0.257 – the worst score – in (c)). Because these two subjects score near chance under three independently trained and validated pipelines (a hand-engineered single-subject model, a hand-engineered LOSO model, and an end-to-end deep network), the most plausible explanation is a genuine, subject-intrinsic limitation in the recorded signal quality or the strength of that individual’s motor-imagery-related cortical response – rather than a

shortcoming specific to any one feature-extraction or classification method.

5.5.3 General Performance Pattern

A subtler pattern emerges when comparing the *spread* of the bars, not just their mean height: the standard deviation across subjects actually *increases* from panel (a) to (c) ($0.074 \rightarrow 0.125 \rightarrow 0.140$) even as the mean improves. In other words, the accuracy gains achieved by moving from single-subject training, to LOSO, to EEGNet are *not* shared equally by every subject: subjects that were already relatively decodable under the weakest pipeline (S01, S03, S08, S09) gain the most as stronger pipelines are introduced, while the hardest subjects (S02, S05, and to a lesser extent S06) remain anchored close to the 0.25 chance line across all three charts, essentially unmoved by either LOSO pooling or the switch to EEGNet. This visually reinforces the conclusion drawn from the pairwise-transfer analysis (Section 5.2, Figure 5): part of the cross-subject accuracy gap in this dataset reflects genuine, hard-to-close differences in signal quality between individuals, rather than a limitation that better feature engineering or a better classifier alone can fully resolve.

6 Discussion

6.1 Domain Shift Across Subjects

The pairwise transfer-accuracy heatmap (Figure 5) provides the clearest quantitative picture of the inter-subject domain-shift problem in this dataset: within-subject decoding accuracy (0.656 mean) is more than two-and-a-half times higher than direct, unadapted cross-subject transfer accuracy (0.254 mean, statistically at chance). This gap arises because CSP spatial filters are, by construction, optimized to the specific spatial covariance structure of the class-conditional signal for the subject on which they were fit; that covariance structure is shaped by individual factors – skull/scalp conductivity, cortical folding, exact electrode positioning relative to the sensorimotor strip, and the subject’s own cognitive strategy for performing motor imagery – none of which are shared across individuals. The WPD statistical features suffer a related, if somewhat less severe, problem: the concentration of signal energy across frequency sub-bands (e.g. where in the mu/beta range a given subject’s ERD is strongest) is itself subject-dependent.

6.2 Why LOSO Training Helps, But Only Modestly

Pooling eight subjects’ data before fitting CSP and the MLP (Experiment 2) forces the spatial filters and classifier decision boundaries to compromise across a range of individual covariance structures rather than over-fitting to just one. This explains the improvement from 0.317 (single-subject training, tested on unseen subjects) to 0.401 (LOSO). However, the improvement is modest because averaging over eight heterogeneous individual covariance structures does not produce spatial filters that are well-matched to *any* of them in particular – it is a compromise solution, not an adapted one. This is consistent with the broader finding in the BCI transfer-learning literature that naive data pooling recovers only part of the gap closed by explicit domain-adaptation techniques (such as the Euclidean Alignment already applied here, or more sophisticated Riemannian-geometry methods).

6.3 Why Deep Representations (EEGNet) Generalize Better

EEGNet’s advantage over the hybrid hand-engineered pipeline, even under a curtailed training schedule, likely stems from several compounding factors:

- **Wider, jointly-learned filter bank.** EEGNet’s temporal convolution learns a bank of $F_1 = 8$ frequency-selective filters directly from the 4–38 Hz data, rather than relying on a single, manually-chosen 8–30 Hz band – it can allocate representational capacity to whichever frequency ranges are most discriminative for the pooled training population, rather than a range chosen *a priori* from domain knowledge.
- **End-to-end joint optimization.** The temporal and spatial filters in EEGNet are learned jointly with the classifier under a single loss function, whereas the hybrid pipeline fits WPD, CSP, and the MLP in three separate, sequential, unsupervised-then-supervised stages – each stage can only pass along information that survived the previous, independently-optimized stage.
- **Strong built-in regularization.** The max-norm weight constraints on both the depthwise-convolution and dense layers act as aggressive implicit regularizers well-suited to the modest amount of per-subject training data available, likely improving generalization more than the dropout-only regularization used in the MLP.
- **Per-trial normalization.** Z-scoring each trial independently (rather than using a training-set-derived global scaler, as in the hybrid pipeline) directly removes inter-subject amplitude/scale differences at the input level, before any learned filtering occurs.

6.4 Impact of the Computational Constraint on the EEGNet Results

The results in Section 5.3 should be interpreted as a **conservative lower bound** on EEGNet’s true cross-subject performance on this dataset. With only 42 epochs of budget (versus a recommended 300) and a correspondingly shortened early-stopping patience (6 vs. 30), several folds (e.g. S01, S02) had not yet plateaued in validation accuracy when the epoch budget was exhausted – they were not stopped early by the patience criterion, but simply ran out of allotted epochs. This strongly suggests that a fully-trained EEGNet, given the estimated 7+ hours of additional CPU time that were not available for this project, would likely further widen its already-observed advantage over the hand-engineered baselines, consistent with published EEGNet benchmarks on BCI-IV-2a that typically report higher LOSO accuracies under full training budgets and (in some cases) GPU-accelerated data augmentation.

6.5 Limitations

- Only the T (training) session per subject was usable for EEGNet due to missing external label files for the E (evaluation) sessions, halving the amount of data theoretically available in the original BCI-IV-2a competition protocol.
- The EEGNet training budget was constrained by CPU-only compute, as discussed above; results for EEGNet should not yet be read as reflecting its ceiling performance on this dataset.
- The three experiments use two different pre-processing pipelines with different band-pass ranges, epoch windows, and resampling rates, which somewhat complicates a perfectly like-for-like comparison between the hybrid and deep-learning approaches – however, this reflects a realistic comparison of complete, self-consistent competing pipelines, not an ablation of a single variable.

- With only nine subjects, per-subject accuracy estimates (and their standard deviations) are based on limited sample sizes; some subject-to-subject differences reported here may not be statistically robust to a larger cohort.

7 Future Work

- **Fully-trained EEGNet.** Re-run the LOSO EEGNet experiment with the originally intended configuration (300 epochs, patience 30, data augmentation enabled) on GPU-accelerated hardware to obtain an unconstrained estimate of the architecture’s ceiling performance on this dataset, and to properly benefit from the noise-injection and time-shift augmentation strategies already implemented but disabled in this run.
- **Transfer learning and calibration.** The fine-tuning experiment (Section 5.3.1) already demonstrates the value of brief subject-specific calibration; this could be extended with few-shot / meta-learning approaches (e.g. model-agnostic meta-learning) explicitly optimized to adapt quickly from very small calibration sets, or with domain-adversarial training that encourages subject-invariant intermediate representations during the main LOSO training phase itself.
- **Riemannian-geometry methods.** Beyond Euclidean Alignment, Riemannian tangent-space mapping of trial covariance matrices is a well-established family of techniques for cross-subject BCI transfer that could be integrated with either the hybrid CSP pipeline or as a pre-processing step ahead of EEGNet.
- **Larger and more diverse cohorts.** Validating these findings on additional public motor-imagery datasets (e.g. Physionet EEGMMI, OpenBMI) with larger subject pools would clarify how robust the observed generalization ordering (subject-dependent < LOSO hybrid < LOSO EEGNet < fine-tuned EEGNet) is across populations and recording protocols.
- **Ensembling and hybridization.** Combining the CSP/WPD hand-engineered features with EEGNet’s learned representations (e.g. as parallel input streams into a joint classifier, or via late-fusion ensembling) may capture complementary information and further improve cross-subject robustness beyond either approach alone.
- **Using both T and E sessions.** Recovering or re-deriving the true labels for the evaluation (E) sessions would double the amount of per-subject data available for training and calibration, which is likely to particularly benefit the data-hungry EEGNet model.

8 Conclusion

This report presented a systematic, three-way comparison of cross-subject generalization strategies for EEG motor-imagery classification on the nine-subject BCI Competition IV-2a dataset. A naively-trained, single-subject hybrid WPD-CSP-MLP pipeline achieves strong in-sample accuracy (0.759) but generalizes poorly to unseen subjects (0.317 ± 0.074 , close to chance for several individuals), directly quantifying the severity of inter-subject domain shift in EEG-based BCIs. A Leave-One-Subject-Out cross-validation protocol applied to the same feature family, augmented with Euclidean Alignment, recovers a modest improvement to 0.401 ± 0.125 . Replacing the hand-engineered feature pipeline with EEGNet, an end-to-end deep convolutional

architecture, and evaluating under the identical LOSO protocol, achieves the best cross-subject performance of the three approaches (0.439 ± 0.140 , Cohen’s $\kappa = 0.252$) despite operating under a substantially reduced training budget necessitated by CPU-only compute – and this figure improves further to 0.476 with a brief, practical subject-specific fine-tuning step. Taken together, these findings support two central conclusions relevant to the design of subject-independent BCIs: first, that naive pooling of multi-subject training data (LOSO) only partially addresses the inter-subject domain-shift problem inherent to hand-engineered spatial filters such as CSP; and second, that learned, end-to-end deep representations such as EEGNet – even when computationally constrained – transfer across subjects more robustly than hand-engineered features, and that inexpensive per-subject calibration remains a highly effective complementary strategy for closing the remaining gap.

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- [9] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, et al., “PyTorch: An imperative style, high-performance deep learning library,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019.

A Additional Figures

This appendix collects supplementary diagnostic figures referenced in the main text but not central to the primary results narrative.

A.1 Experiment 1 – Supplementary Figures

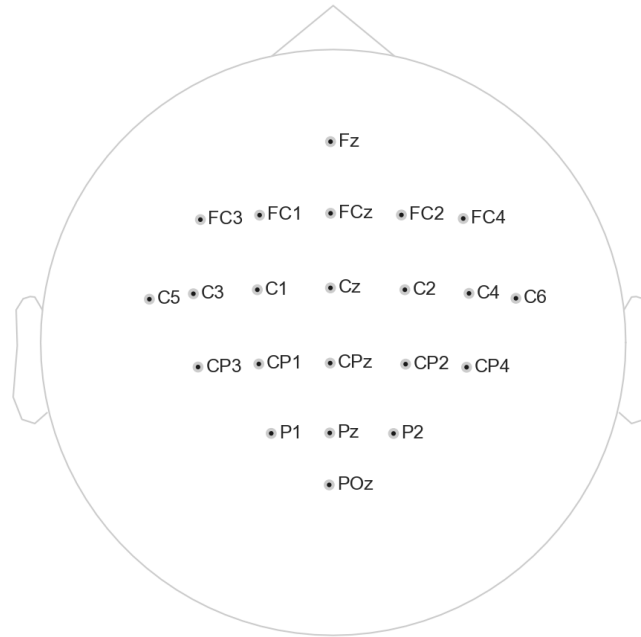


Figure 8: 10–20 electrode layout used for Subject A01T recordings, showing the 22 EEG and 3 EOG sensor positions after channel renaming.

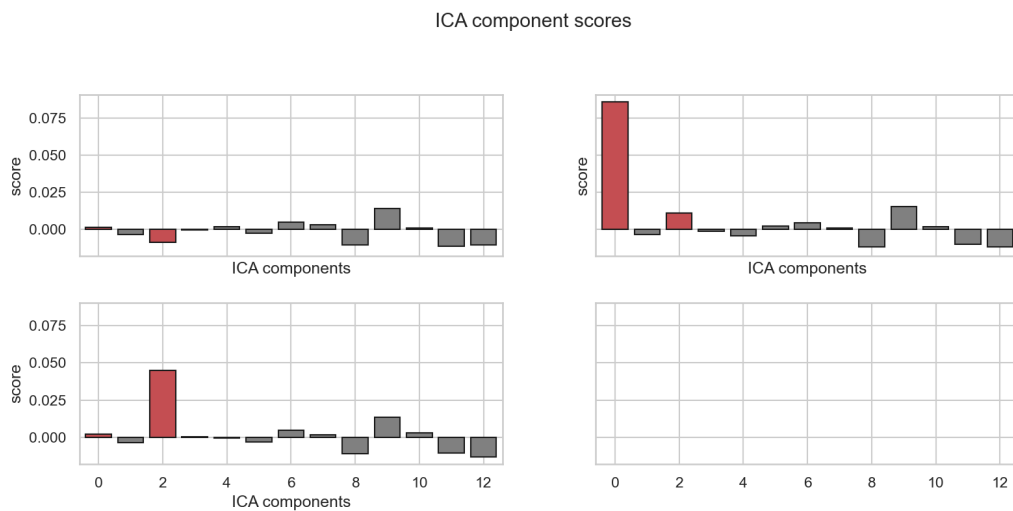


Figure 9: ICA component–EOG correlation scores used to automatically flag ocular-artifact components for removal (threshold = 2.5), Subject A01T. Flagged components are excluded from the reconstructed signal prior to band-pass filtering.

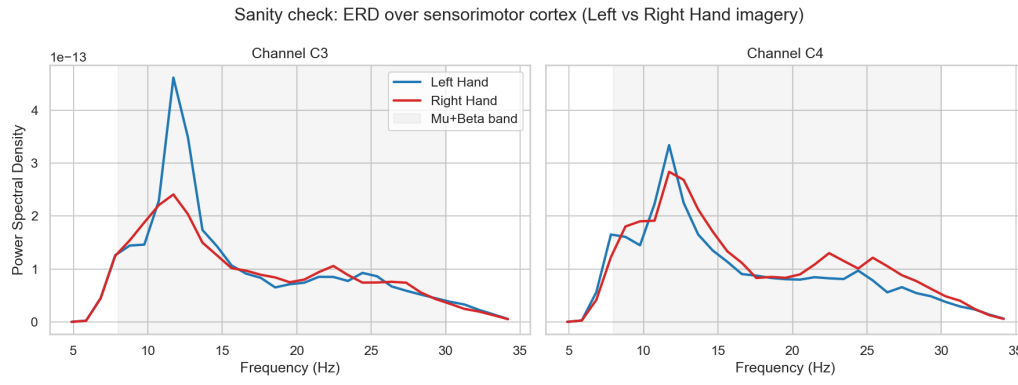


Figure 10: Sanity check for event-related desynchronization (ERD): mean power spectral density at channels C3 and C4 for left- vs. right-hand imagery, Subject A01T. The expected contralateral ERD pattern is visible within the shaded mu+beta band.

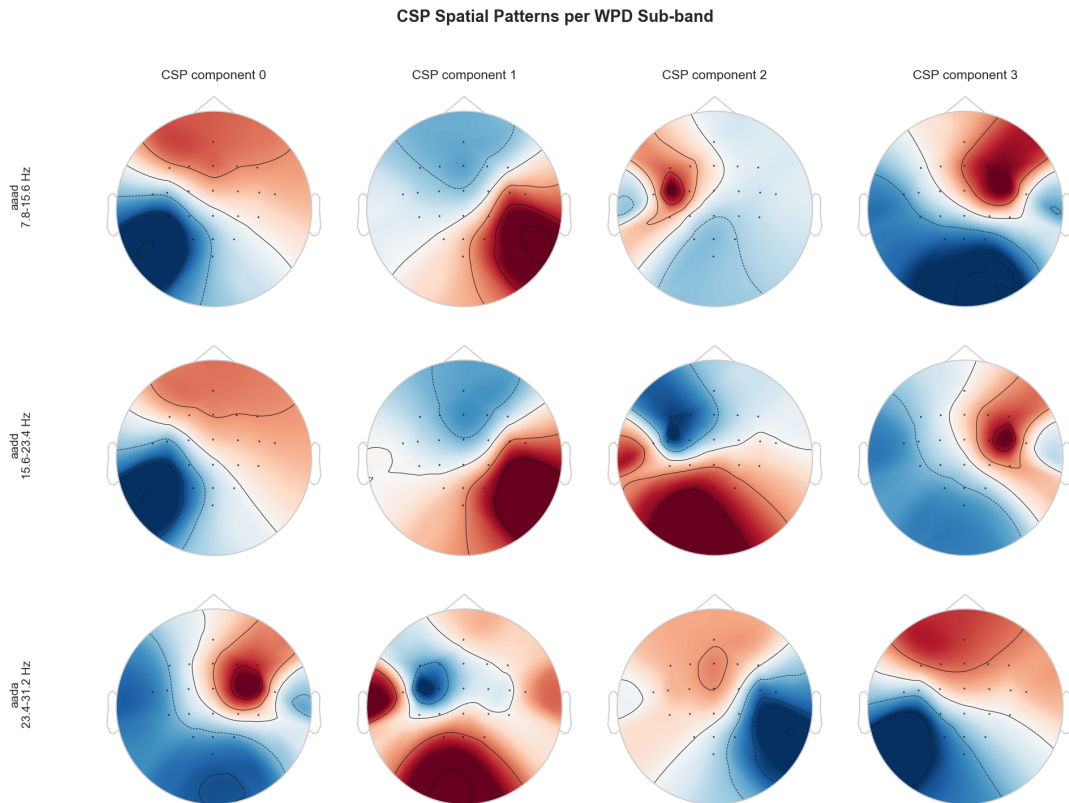


Figure 11: CSP spatial-filter topographies for each of the three relevant WPD sub-bands (rows) and the top four spatial components (columns), Subject A01T.



Figure 12: Training and validation loss/accuracy curves for the subject-dependent MLP (Subject A01T only). Early stopping halted training at epoch 65; the lowest-validation-loss weights were restored.

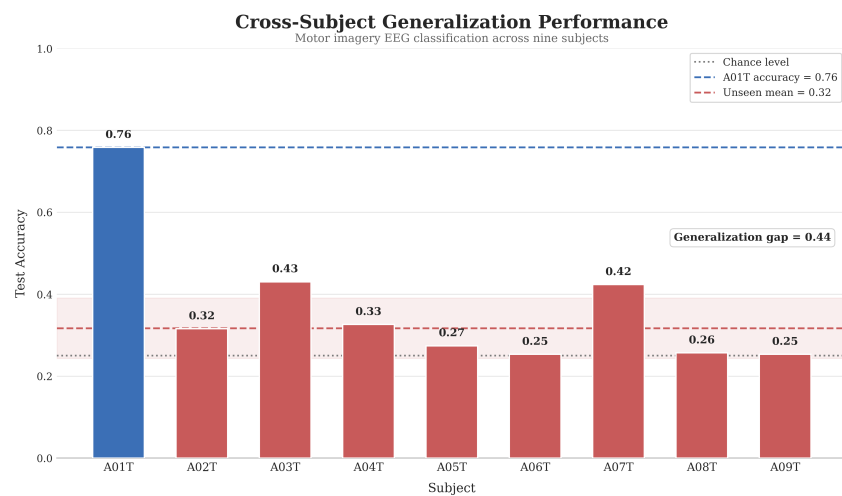


Figure 13: Overall cross-subject generalization summary: per-subject accuracy with reference lines for chance level, in-sample (A01T) accuracy, and mean unseen-subject accuracy; the shaded band indicates ± 1 SD across unseen subjects.

A.2 Experiment 2 – Supplementary Figures

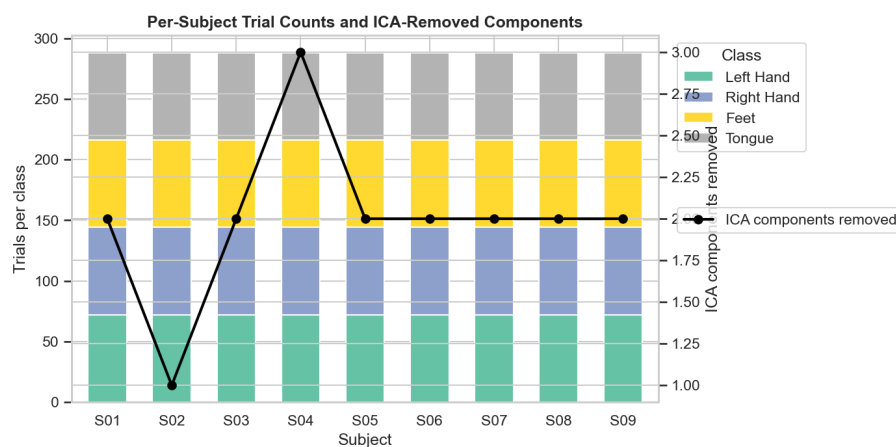


Figure 14: Per-subject trial counts by class (stacked bars) and number of ICA components removed as ocular artifacts (line), across all nine subjects.

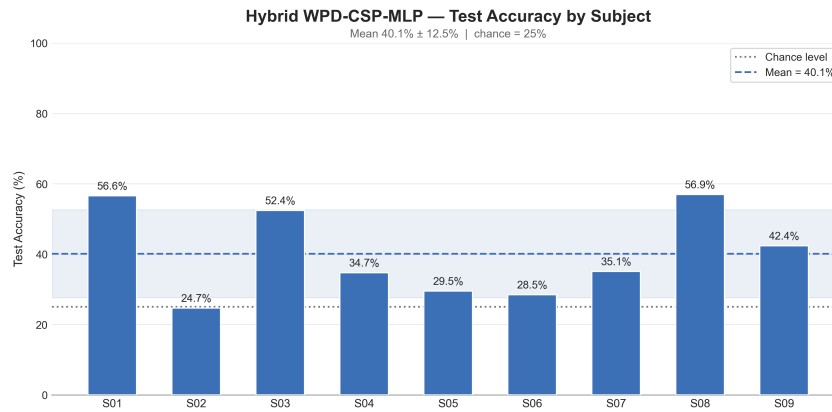


Figure 15: Simplified per-subject mean-accuracy summary for the LOSO Hybrid WPD-CSP-MLP classifier, expressed as a percentage, without the baseline or per-class breakdown shown in Figure 3.

Table 11: Experiment 2 – full per-class F1-score breakdown across all nine LOSO folds.

Test Subject	F1 (Left Hand)	F1 (Right Hand)	F1 (Feet)	F1 (Tongue)
S01	0.634	0.680	0.581	0.150
S02	0.000	0.218	0.382	0.027
S03	0.576	0.442	0.465	0.532
S04	0.286	0.419	0.026	0.443
S05	0.345	0.388	0.182	0.000
S06	0.053	0.000	0.225	0.453
S07	0.438	0.454	0.301	0.000
S08	0.528	0.639	0.548	0.556
S09	0.482	0.396	0.140	0.500

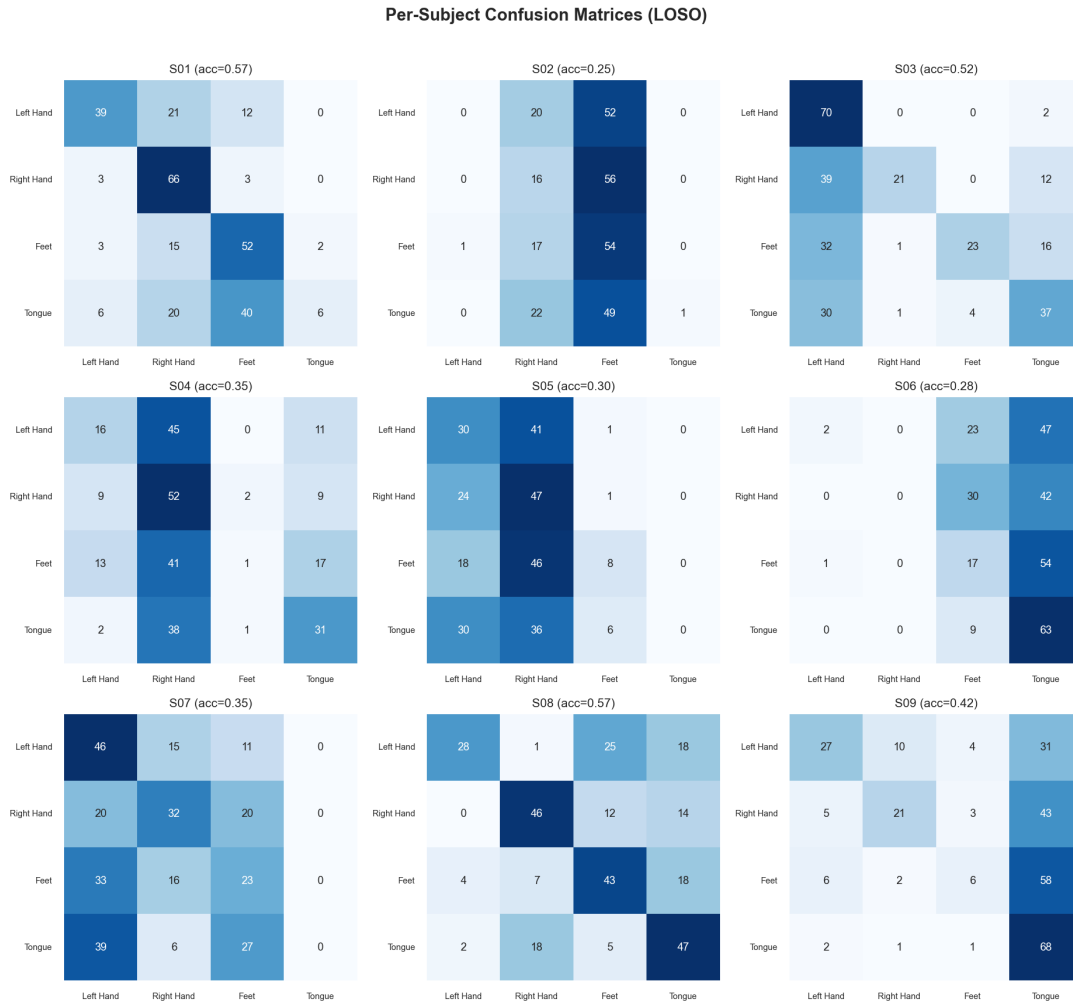


Figure 16: Per-subject confusion matrices for all nine LOSO folds (Hybrid WPD-CSP-MLP).

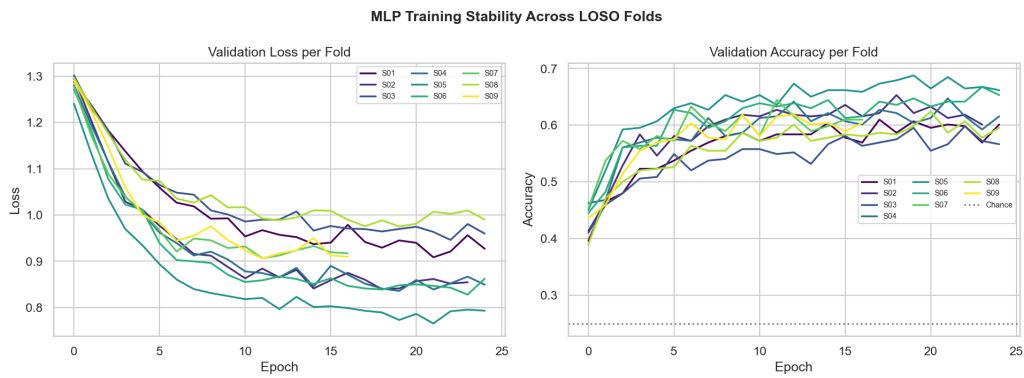


Figure 17: Validation loss and accuracy trajectories for the MLP across all nine LOSO folds, illustrating training stability across subjects.

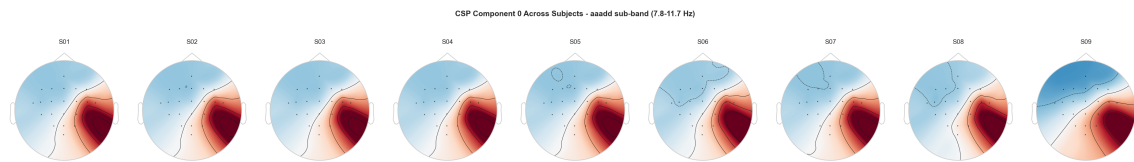


Figure 18: CSP component 0 topography (first relevant sub-band) fitted independently for each subject, illustrating inter-subject variability in the optimal spatial filter – a visual counterpart to the domain-shift result of Figure 5.

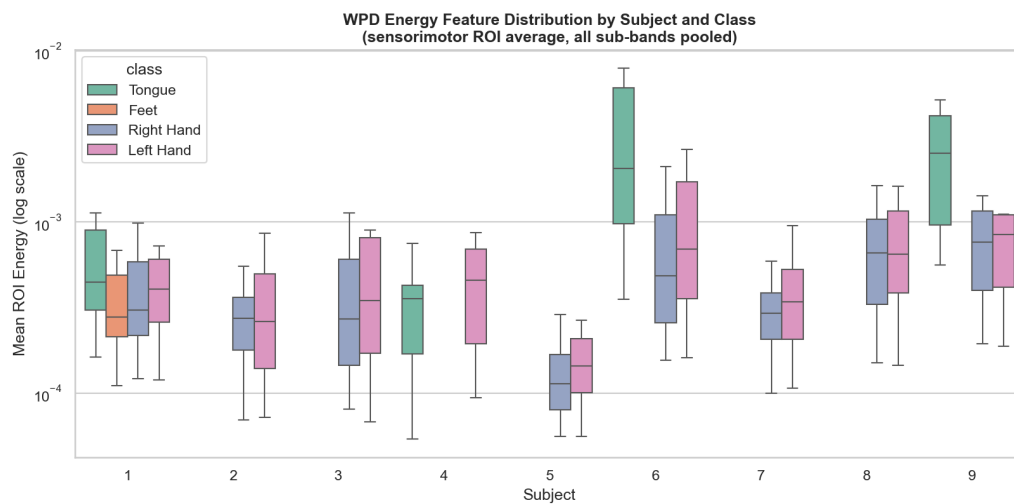


Figure 19: WPD energy feature distribution (sensorimotor ROI average, pooled sub-bands) by subject and class, illustrating heterogeneous energy scales across subjects.

Hybrid WPD-CSP-MLP — Comprehensive LOSO Performance Summary

BCI Competition IV-2a · 9 subjects · 4-class motor imagery (left hand, right hand, feet, tongue)

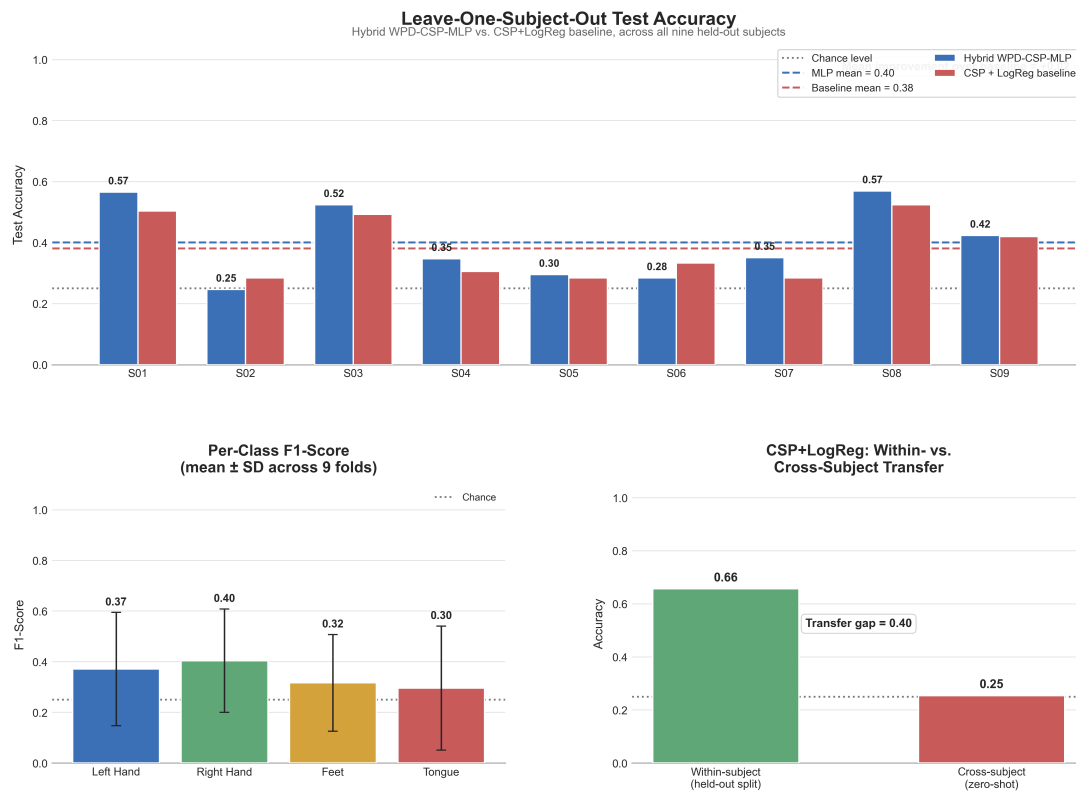


Figure 20: Consolidated LOSO results dashboard summarizing per-subject Hybrid-MLP vs. baseline accuracy alongside auxiliary summary statistics.